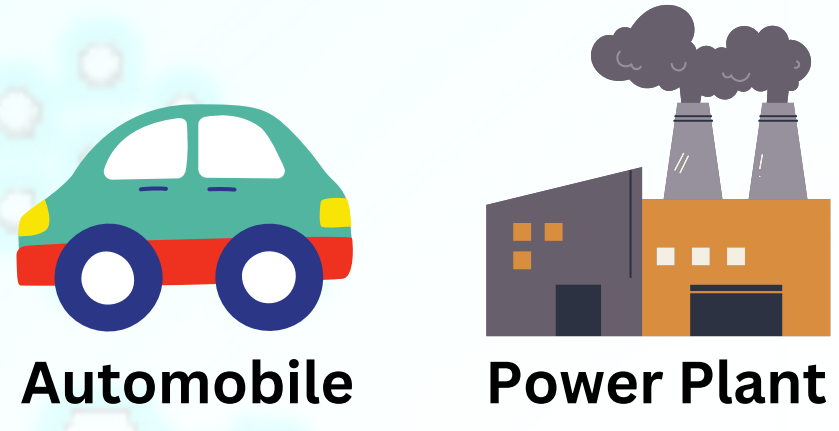


Synergistic Combination of $\text{Sb}_2\text{Si}_2\text{Te}_6$ Additives in $\text{Bi}_{0.4}\text{Sb}_{1.6}\text{Te}_3$ for Enhanced Thermoelectric Waste Heat Harvesting

EY007

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INTRODUCTION

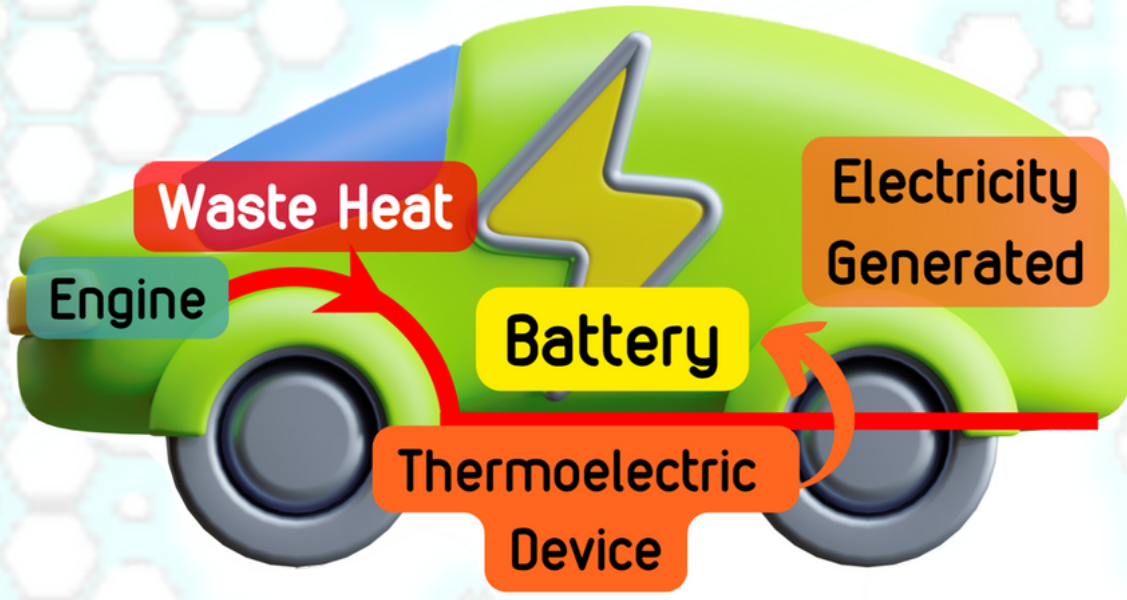


Problem: Waste Heat

- 66% of energy inputs are wasted as heat
- Contributes to global warming

Solution: Thermoelectrics

APPLICATION



Problem with current thermoelectrics: Low zT

Not cost efficient enough for practical applications

PROBLEM

Efficiency of Thermoelectrics:

$$zT = \frac{S^2 \sigma}{\kappa} T$$

- S : Seebeck coefficient (V/K)
- σ : Electrical conductivity (S/m)
- κ : Thermal conductivity (W/m.K)

Increasing zT is a challenge due to inter-relation between S , σ and κ

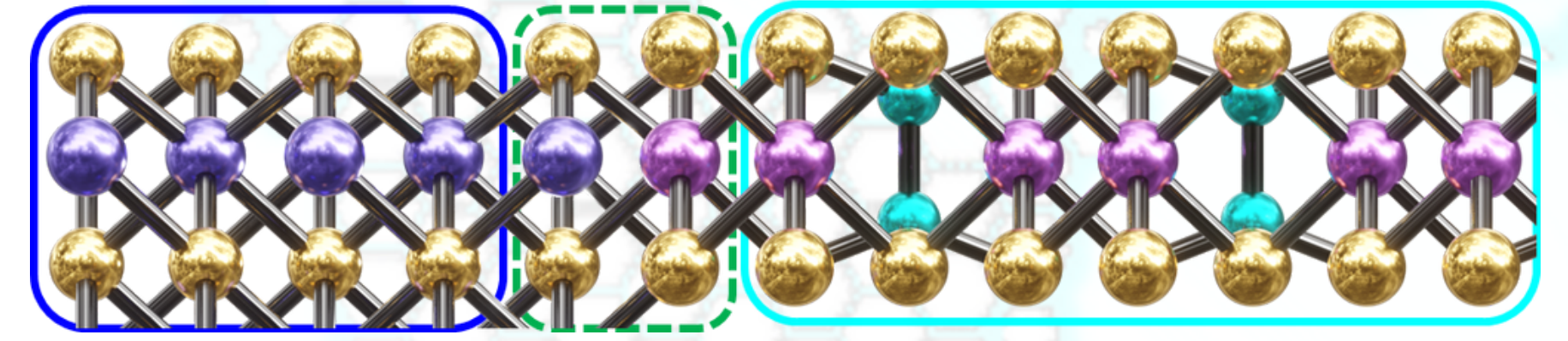
Examples of inter-relations:

- High $\sigma \rightarrow$ High κ
- High $\sigma \rightarrow$ Low $R \rightarrow$ Low $V \rightarrow$ Low S

NOVELTY

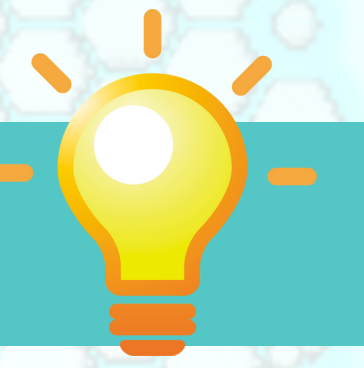
Addition of $\text{Sb}_2\text{Si}_2\text{Te}_6$ Into $\text{Bi}_{0.4}\text{Sb}_{1.6}\text{Te}_3$

$\text{Bi}_{0.4}\text{Sb}_{1.6}\text{Te}_3$ Main Material + $\text{Sb}_2\text{Si}_2\text{Te}_6$ Additive



- High zT near room temperature
- Used in commercially available thermoelectric devices
- Used for testing of novel synthesis methods

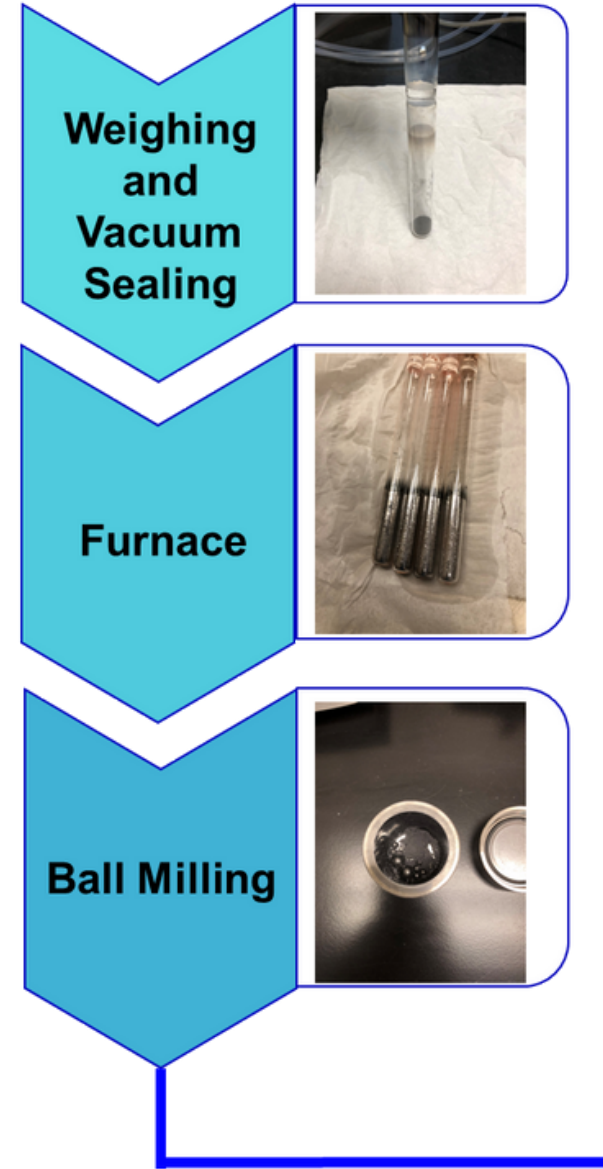
- Low thermal conductivity, forms precipitates $\rightarrow$ Phonon scattering
- Similar structure to $\text{Bi}_{0.4}\text{Sb}_{1.6}\text{Te}_3 \rightarrow$ forms coherent interface
- Higher hole concentration $\rightarrow$ Doping $\rightarrow$ Increases σ , Decreases κ



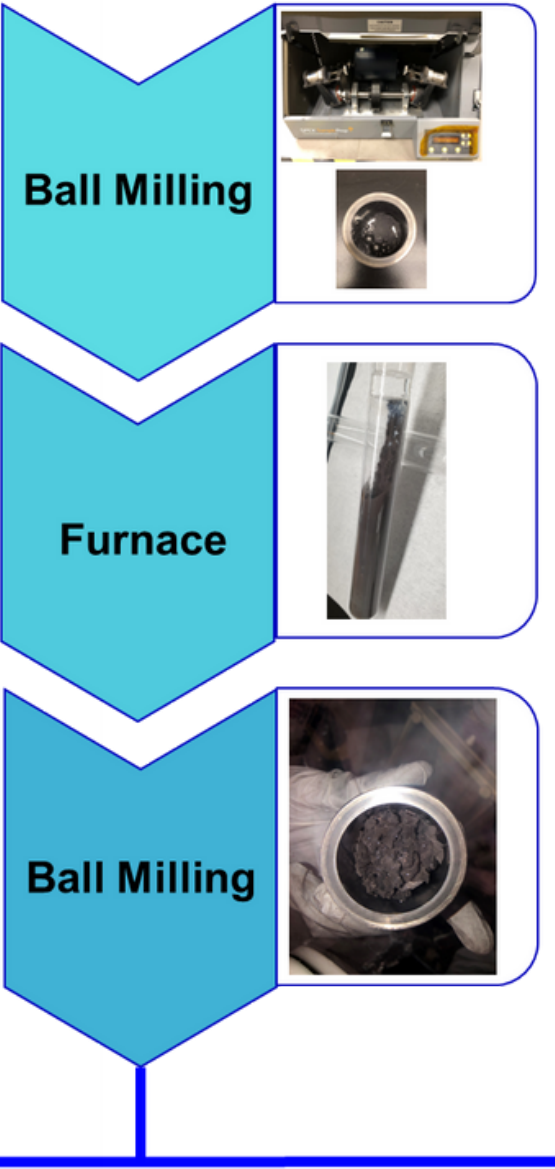
METHODOLOGY

$\text{Bi}_{0.4}\text{Sb}_{1.6}\text{Te}_3$ and $\text{Sb}_2\text{Si}_2\text{Te}_6$ powders made separately

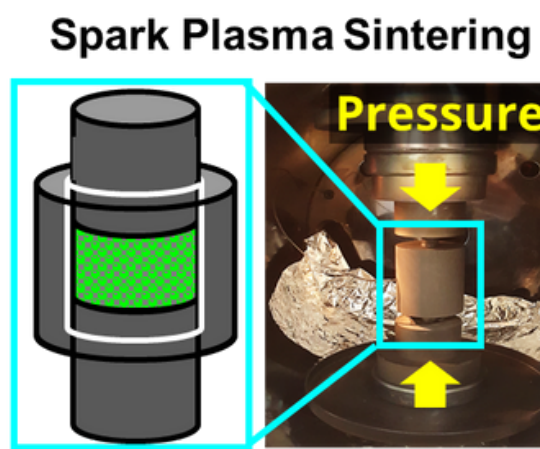
Synthesis of $\text{Bi}_{0.4}\text{Sb}_{1.6}\text{Te}_3$:



Synthesis of $\text{Sb}_2\text{Si}_2\text{Te}_6$:



Mixing to produce 6 samples: $\text{Bi}_{0.4}\text{Sb}_{1.6}\text{Te}_3 + x \text{ mol\% } \text{Sb}_2\text{Si}_2\text{Te}_6$ $x = 0, 0.5, 1, 2, 3$ and 4%.



Analysis

$\text{Sb}_2\text{Si}_2\text{Te}_6$ PRESENT AS PRECIPITATES

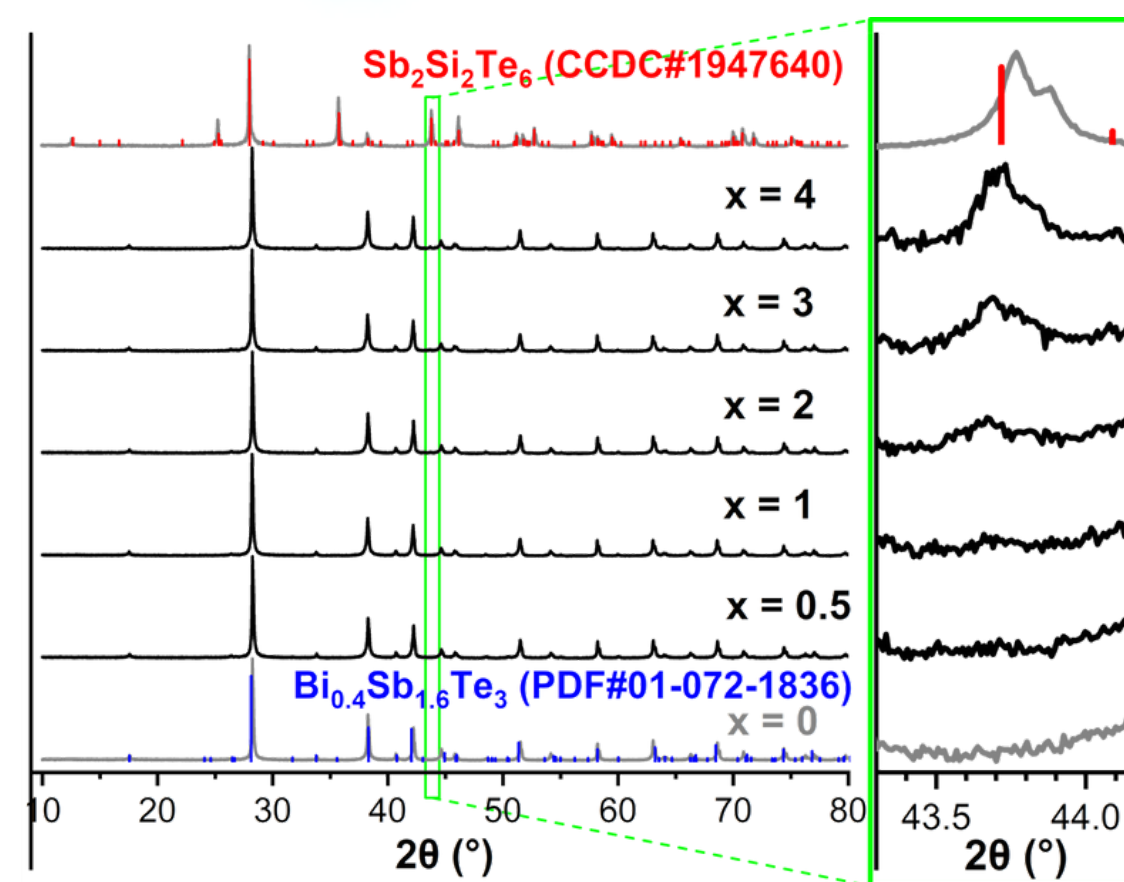


Figure 1: X-ray diffraction (XRD) of the 6 samples

From XRD: Impurity peaks observed, corresponds to $\text{Sb}_2\text{Si}_2\text{Te}_6$ fingerprint

Nano precipitates are $\text{Sb}_2\text{Si}_2\text{Te}_6 \rightarrow$ Phonon scattering, reducing κ

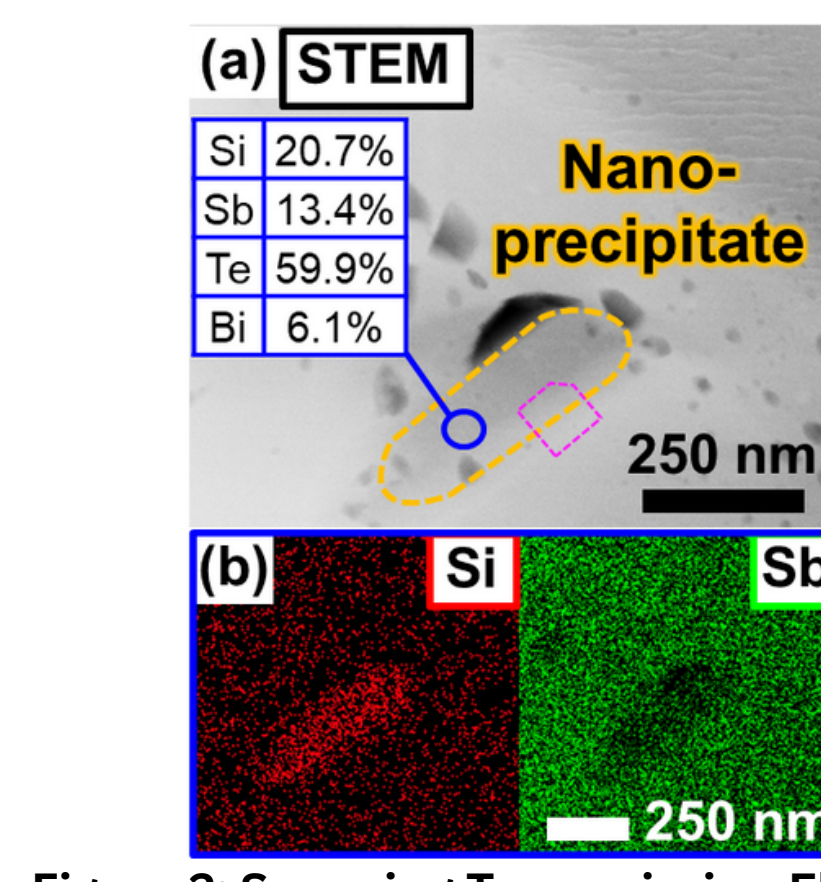


Figure 2: Scanning Transmission Electron Microscope image of the $x = 1\%$ sample

From STEM: Nano precipitates contain Si

WHY $\text{Sb}_2\text{Si}_2\text{Te}_6$ AS ADDITIVE?

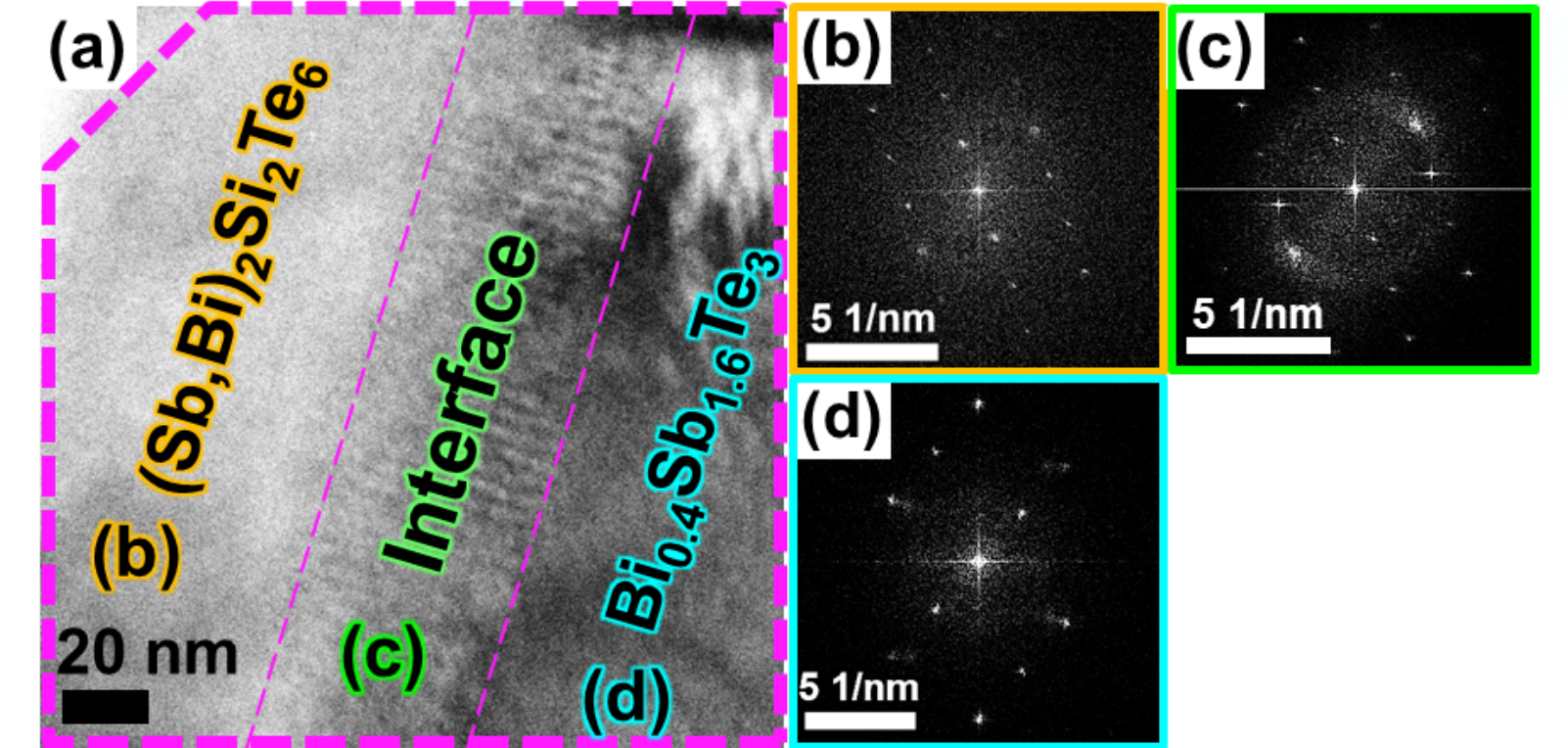


Figure 3: (a) High-Resolution Transmission Electron Microscope image of the $x = 1\%$ sample. Fast Fourier Transformation patterns for (b) $(\text{Sb}, \text{Bi})_2\text{Si}_2\text{Te}_6$, (c) Interface, (d) $\text{Bi}_{0.4}\text{Sb}_{1.6}\text{Te}_3$ phases

Electron Diffraction Pattern: Discrete Dots $\rightarrow$ Highly ordered and crystalline structure

Good carrier mobility between phases

ELECTRICAL PROPERTIES

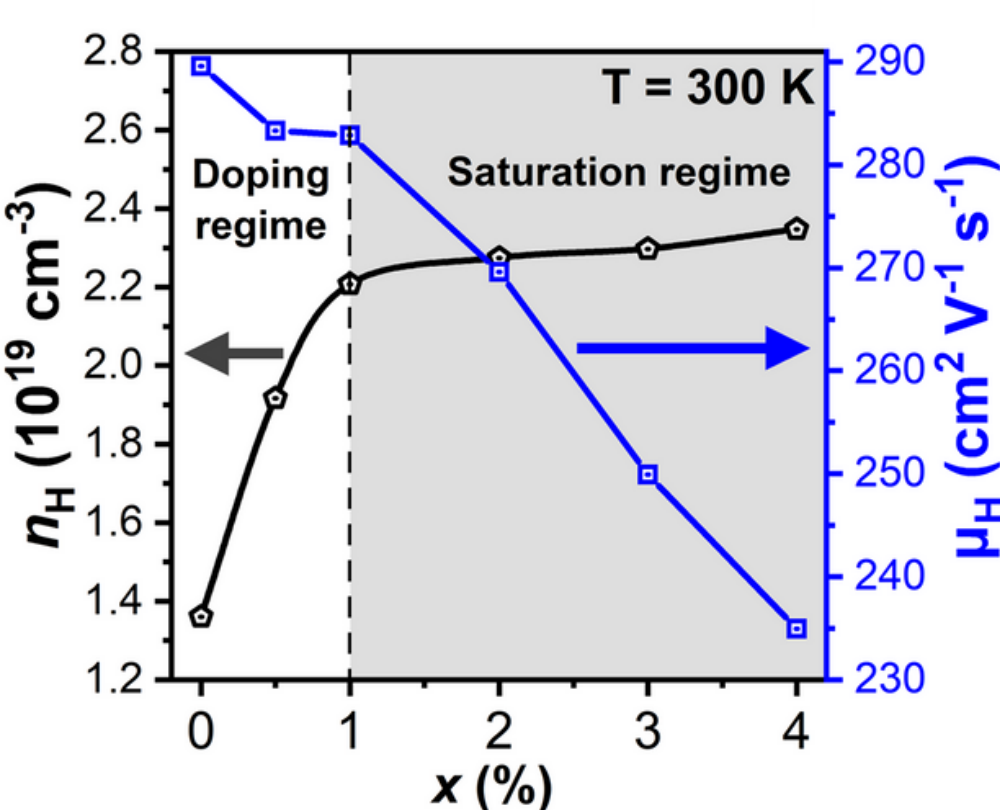


Figure 3a: Hole concentrations (n_H) and hole mobilities (μ_H) at room temperature.

Hole concentration (n_H): $\text{Sb}_2\text{Si}_2\text{Te}_6$ higher

- $x = 0\% - 1\%$: n increases
- $x = 1\% - 4\%$: n saturated

Hole mobility (μ_H): $\text{Bi}_{0.4}\text{Sb}_{1.6}\text{Te}_3$ higher

- $x = 0\% - 1\%$: small drop in μ (Ordered interface)

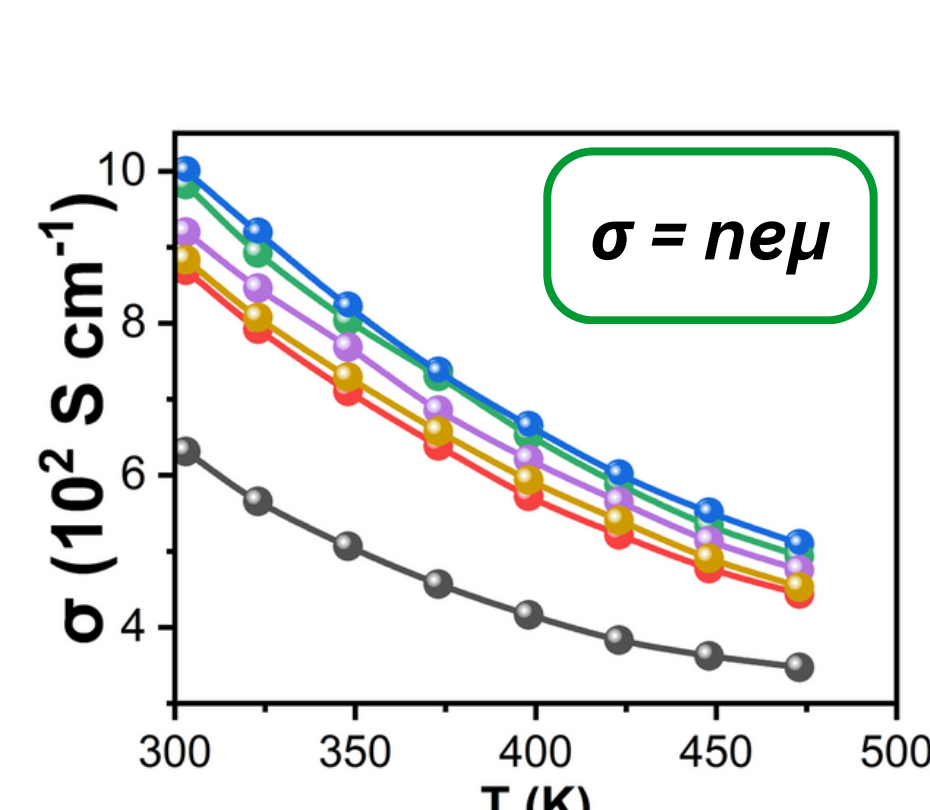


Figure 3b: Electrical conductivity of all 6 samples

$x = 0\% - 1\%$: σ increases

- Increase in n , small drop in μ

$x = 1\% - 4\%$: σ decreases

- Saturated n , large drop in μ

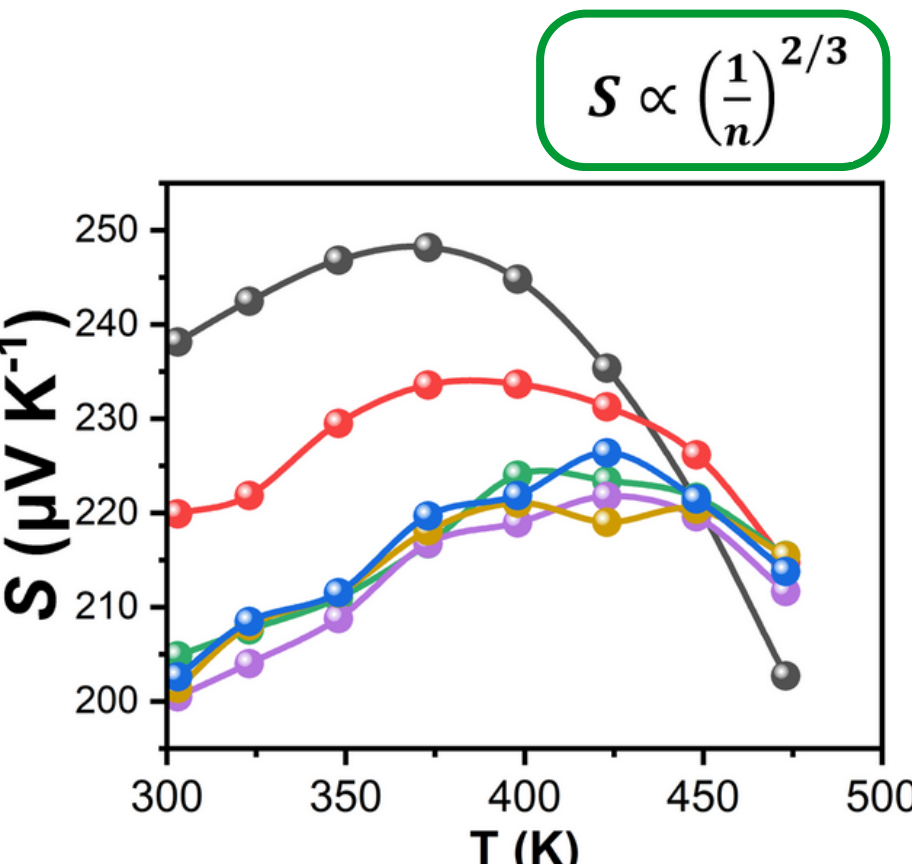


Figure 3c: Seebeck coefficient of all 6 samples

$S \propto \left(\frac{1}{n}\right)^{2/3}$

$x = 0\% - 1\%$: S decreases

- Increase in n

$x = 1\% - 4\%$: S constant

- Saturated n

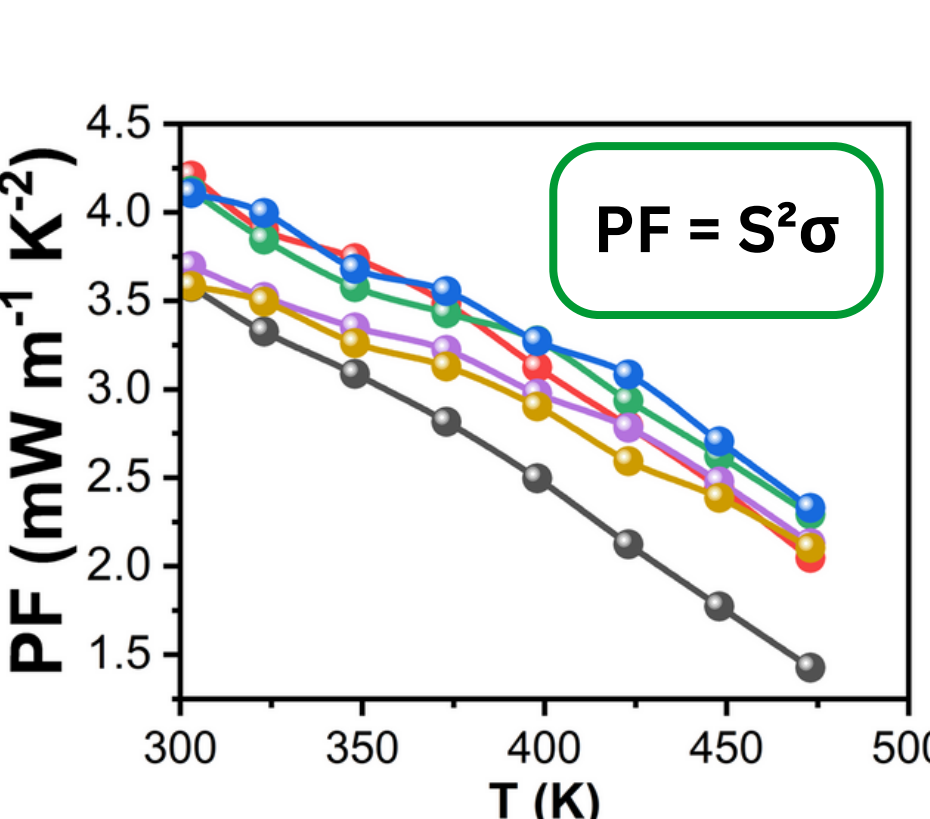


Figure 3d: Power Factor of all 6 samples

$x = 0\% - 1\%$: PF increases

- Optimal trade-offs between n and μ

$x = 1\% - 4\%$: PF decreases

THERMAL PROPERTIES

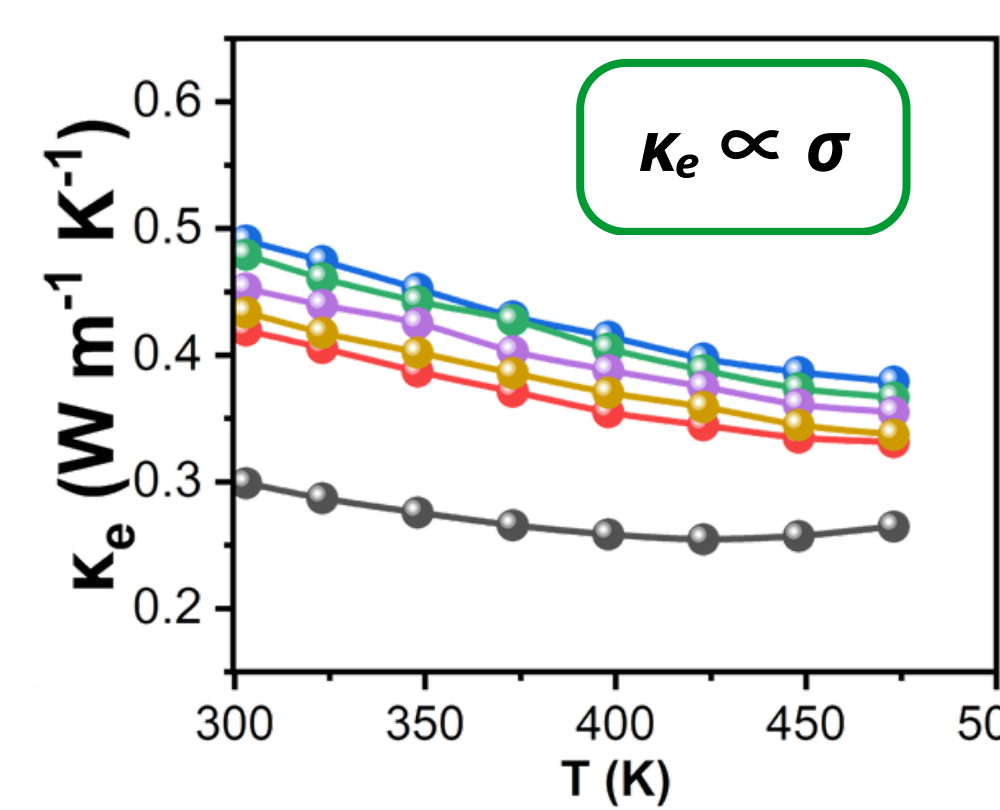


Figure 4a: Electronic contributions of thermal conductivities (κ_e) of all 6 samples

$x = 0.5\% - 4\%$: κ_e increases

- Due to increase in σ

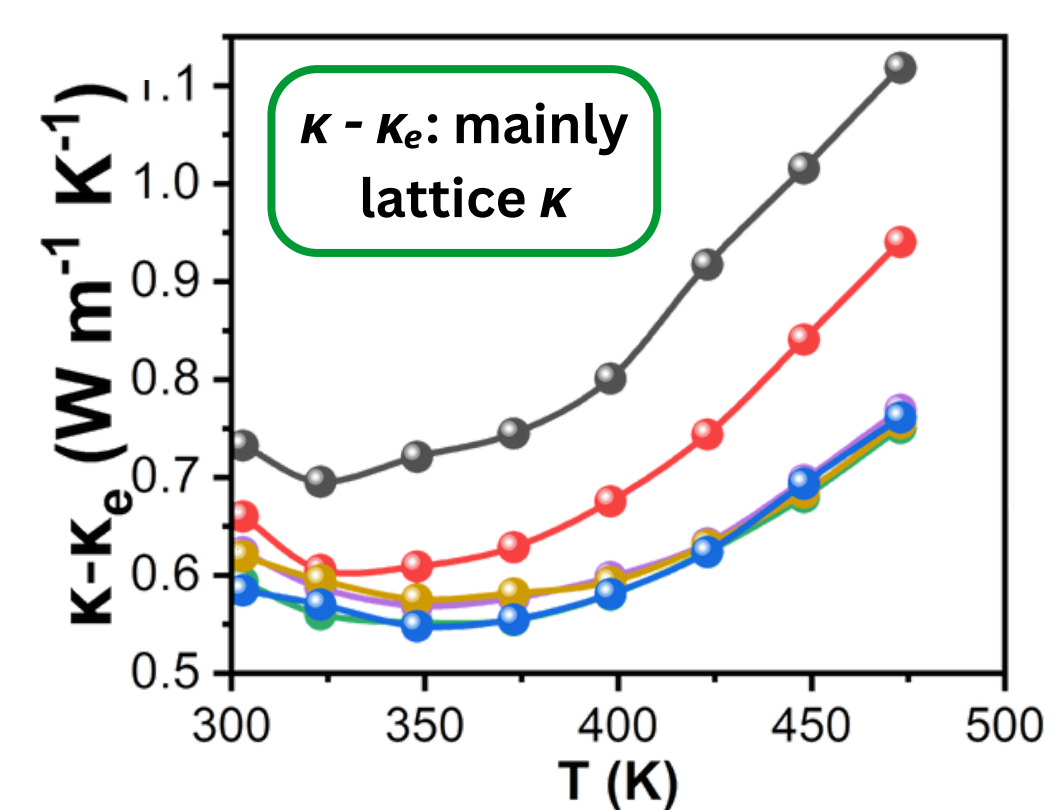


Figure 4b: Lattice and bipolar thermal conductivities ($\kappa - \kappa_e$) of all 6 samples

$x = 0.5\% - 4\%$: $\kappa - \kappa_e$ decreases

- Effective phonon scattering by precipitates

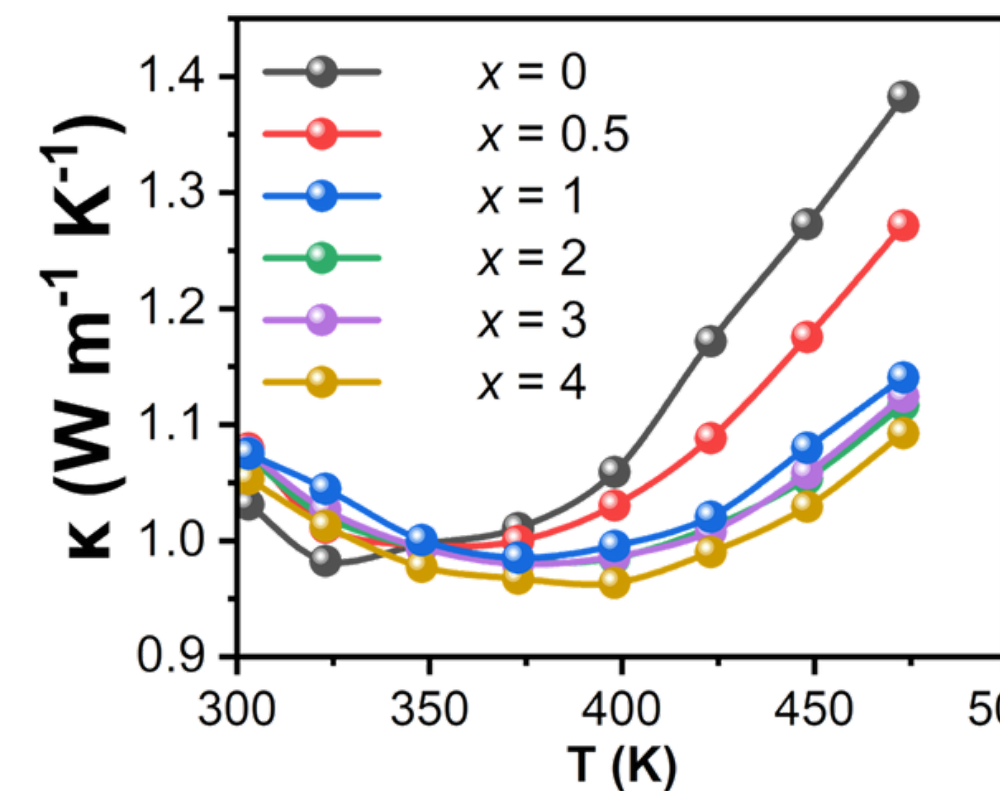


Figure 4c: Total thermal conductivities of all 6 samples

$x = 0\% - 4\%$: κ mostly constant

- Effects of κ_e and $\kappa - \kappa_e$ cancel out

COMPARISON OF RESULTS

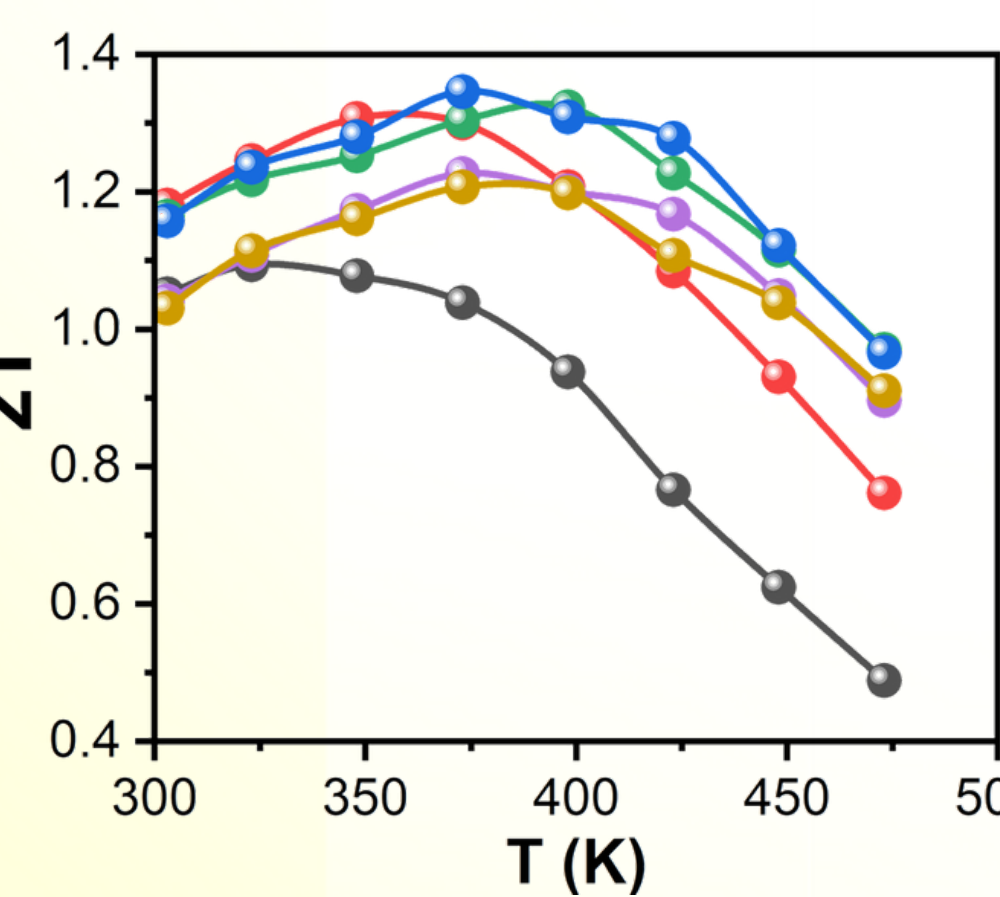


Figure 5a: Figure of merit (zT) of all 6 samples

$x = 1\%$: Highest zT

- 30% increase in peak zT as compared to pristine sample

Increase in x

$x = 4\%$

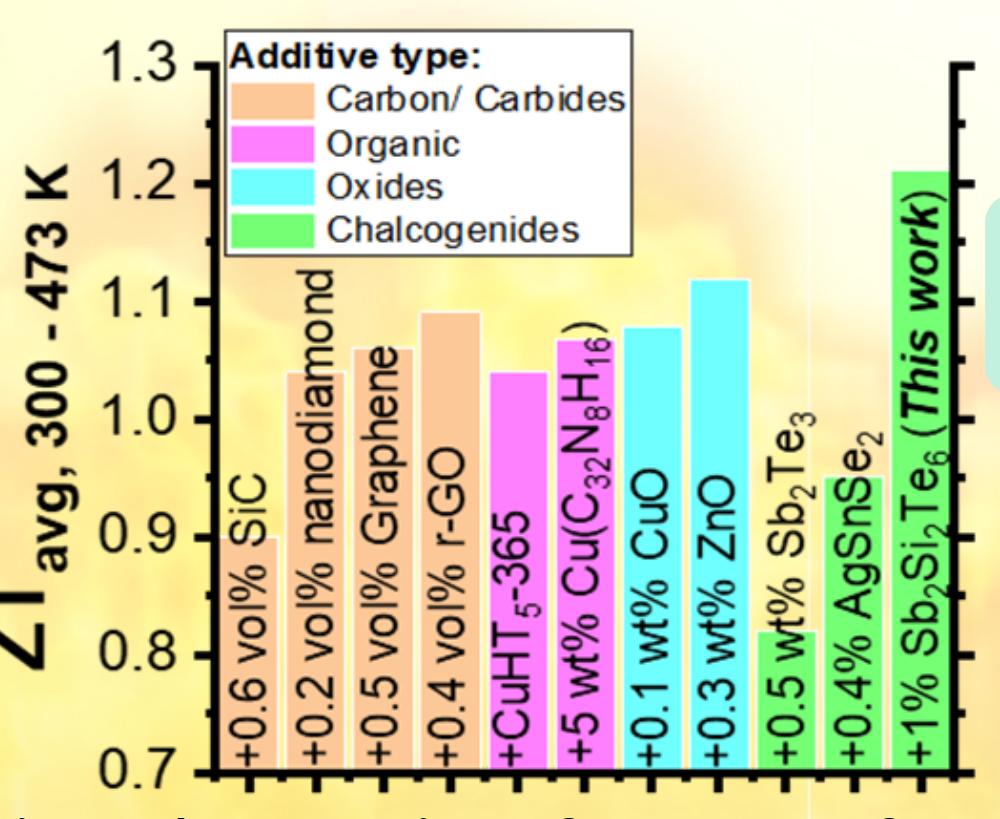


Figure 5b: Comparison of average zT of $x = 1\%$ sample to other BST composites [1-14]

Best average zT as compared to other BST composites

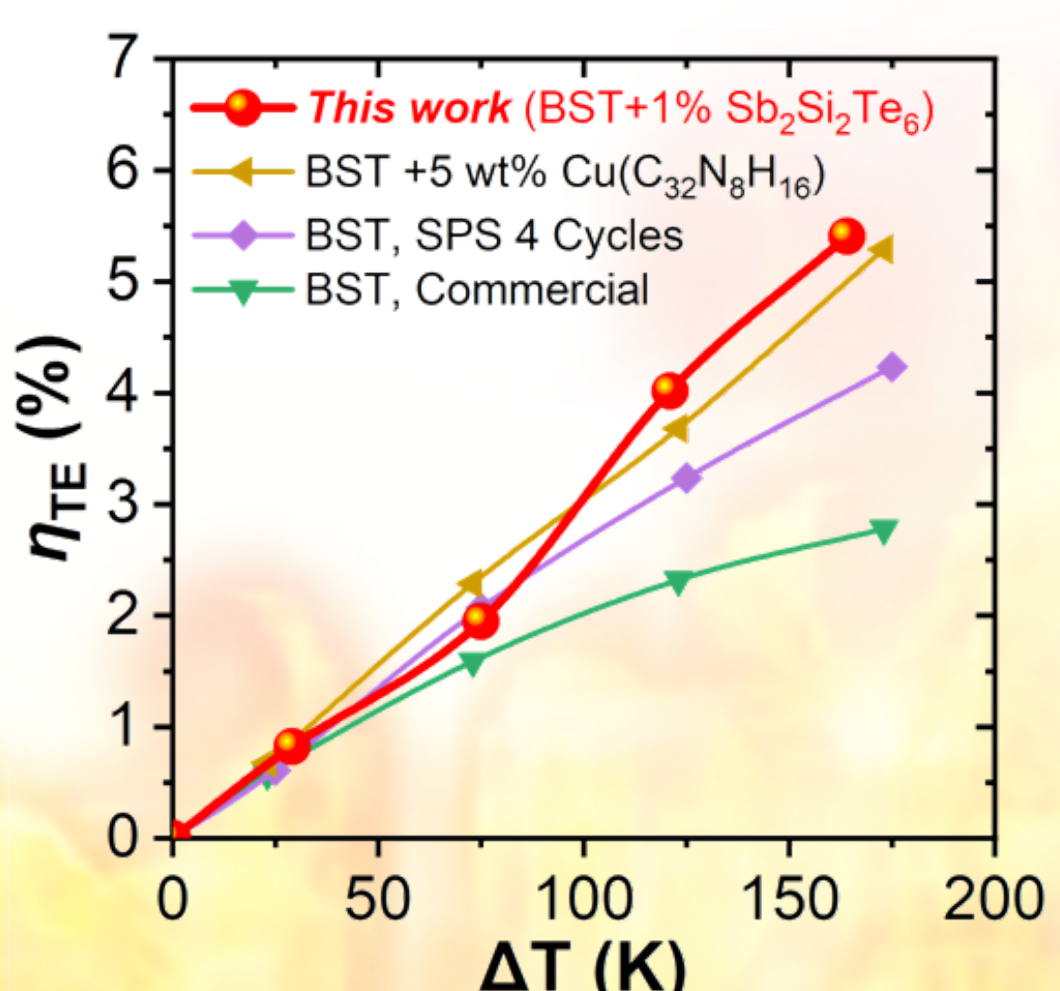


Figure 5c: Thermoelectric power conversion efficiencies (η_{TE}) of $x = 1\%$ sample compared to other p-type $\text{Bi}_{1}\text{Sb}_{2-\gamma}\text{Te}_3$ (BST)

Highest TE efficiency among BST devices

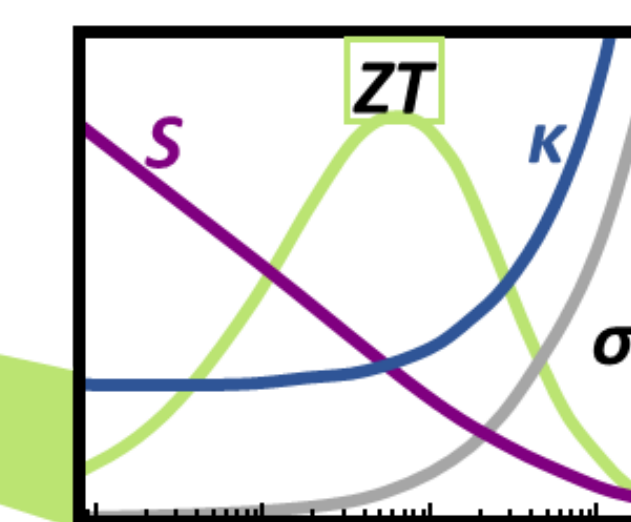
CONCLUSION

- Phonon Scattering
- Coherent interface



Adding 1% $\text{Sb}_2\text{Si}_2\text{Te}_6$

Showcasing synergistic combination between 2 different materials



- Optimal trade-offs between n and μ
- Decrease in $\kappa - \kappa_e$

- Increases zT by 30%
- Best zT_{ave} and η_{TE}

- Improved testing of novel ideas
- Cost-efficient and effective waste heat harvesting.

Combat global warming

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